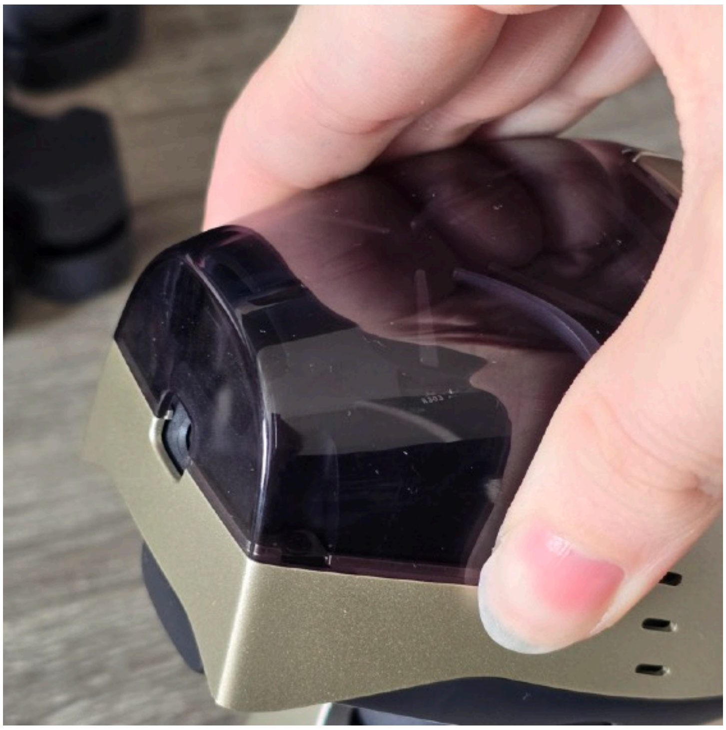
ERS-210 Head Shell Disassembly *by LaikaAibo*

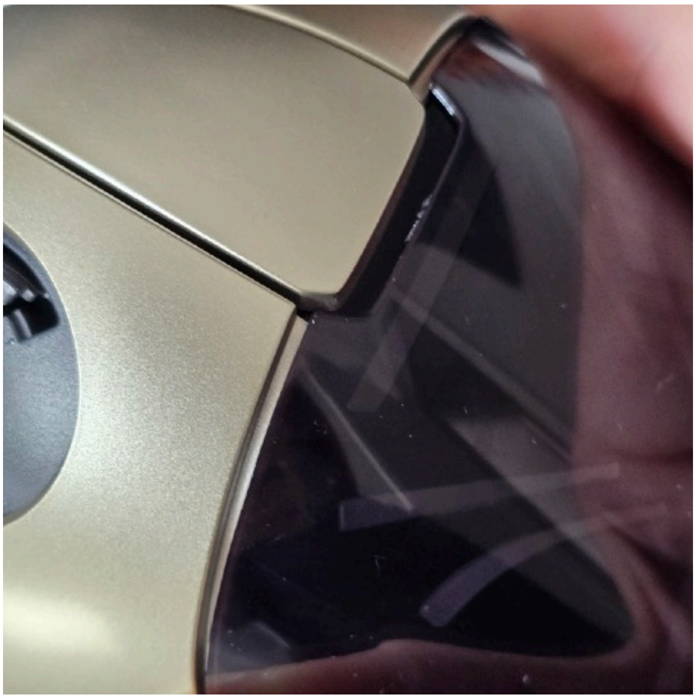
Taking the ERS-210 head apart seems daunting at first, but it is actually straight-forward when guided step-by-step. Please note that you will have to use a bit of force on some steps of the disassembly process, so I recommend reading this guide all of the way through first and then deciding if it's something you are comfortable with doing. Also this is a 20+ year old robot, so a lot of the plastics can be brittle due to aging and storage conditions - proceed with this in mind.

Tools

JIS 0 or 00 screwdriver - I recommend the VESSEL set with the coloured rubber grips for this. Easily purchasable from Amazon.

Before getting started please eject the head module from the body. If you're not sure how to do this, please see the guide on my website. Next we will take both ears off - these can be released by squeezing them inwards near the base and slowly lifting the pegs each side upwards. Once each side is loose, pull directly up, being mindful of the two thinner center pegs. Next we will move onto the visor. Grab each side, squeeze and wiggle.





Eventually you'll start to see gaps appear like this ^
Continue to wiggle, squeeze and pull upwards until there is gaps all away round where the visor joins to the coloured plastic.

The amount of force needed can vary from visor to visor. The front ridge near the nose camera can be the hardest to lift, so don't shy away from using your nail or a prying tool to gently push it upwards to help release the front pegs.

Once all of the pegs release, the visor just pulls up and off, exposing the screws securing the top plastics to the black plastic base.

Next we are going to take off the head button. This is held down by 3 pegs (one at the back and 2 at the front. This section can trip people up as the pegs at the front can easily break if you just try to force the head sensor off. I will show you how to avoid doing that.

We will start with the back peg - simply dig your nail into the gap at this point and lift.

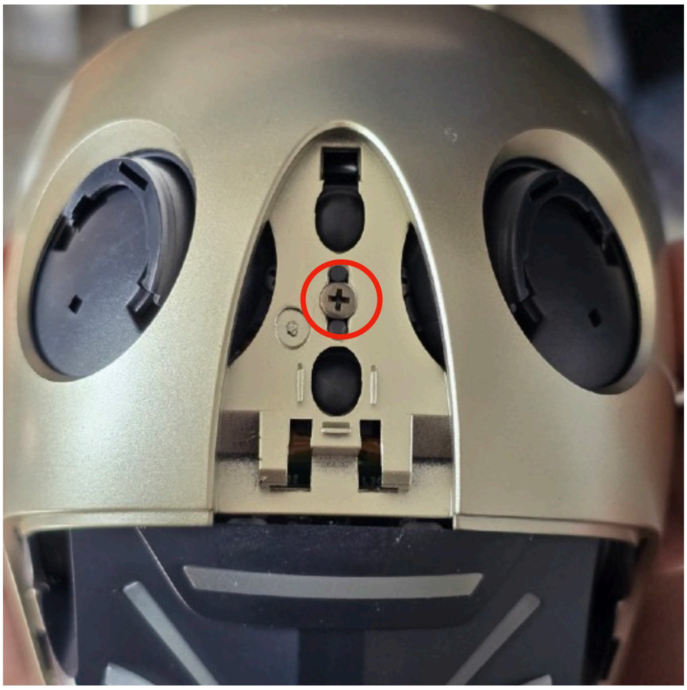
While keeping the back lifted, pull your nail around to the side and push up so you create a sizeable gap like this.

Grab the tiniest flathead from the VESSEL set and insert it into the gap near where the peg on that side is and push up until you feel it release. Take the screwdriver around to the front near the other side and push up, making sure the other two pegs don’t clip back into place.

Once the last peg is released, the head sensor plastic will lift right off along with the spring that gives it it's bounciness. Make sure to keep the spring in a safe place and label it to avoid confusion later on with other springs from the 210.

Here is a close up of the front two pegs to give you a better idea of how they hook in place and the movements needed to release them. ^

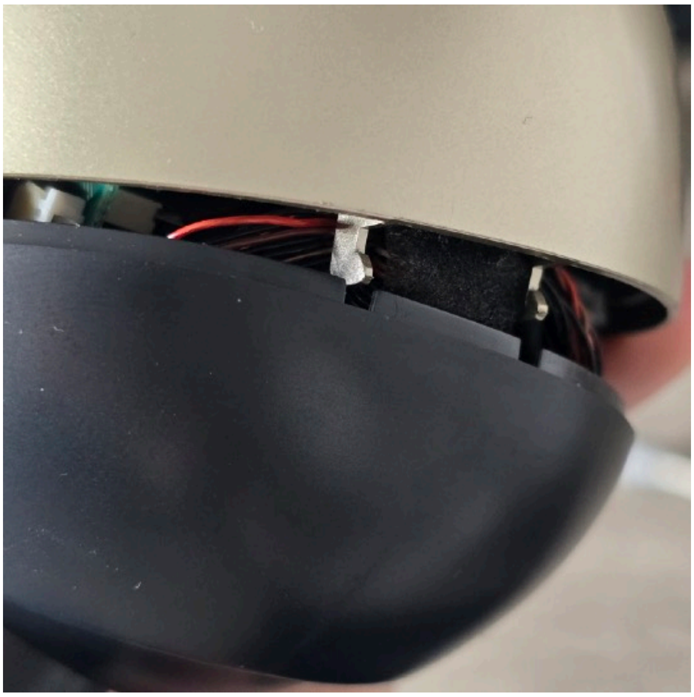
Next we are going to release the coloured plastic and the black eye mask from the lower portion of the head. Take out the 4 screws (two each side) that you can see above.



Then take out this screw in the center.

As soon as those screws are taken out, you'll probably start to see the coloured plastic lifting up like seen above.

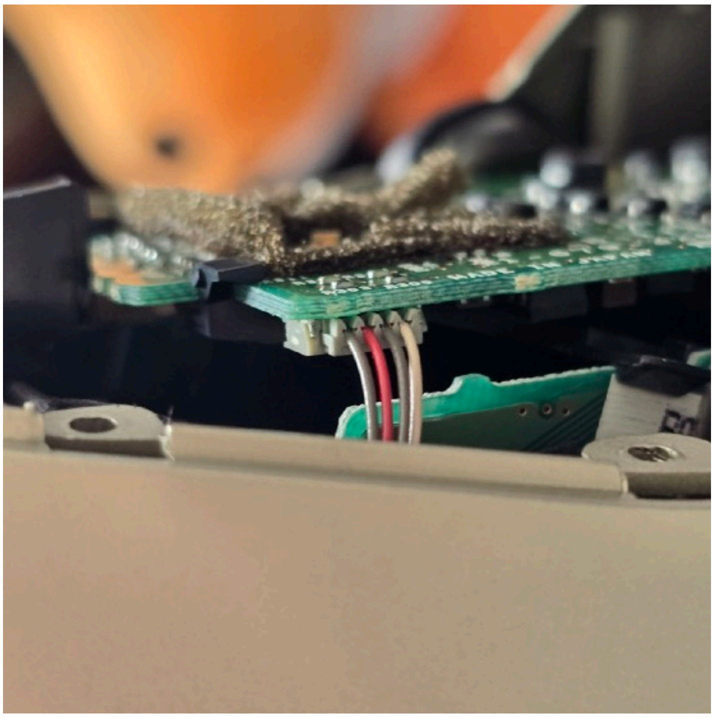
Grab onto the coloured plastic and gently wiggle and lift up until it's lifted up approximately this height ^



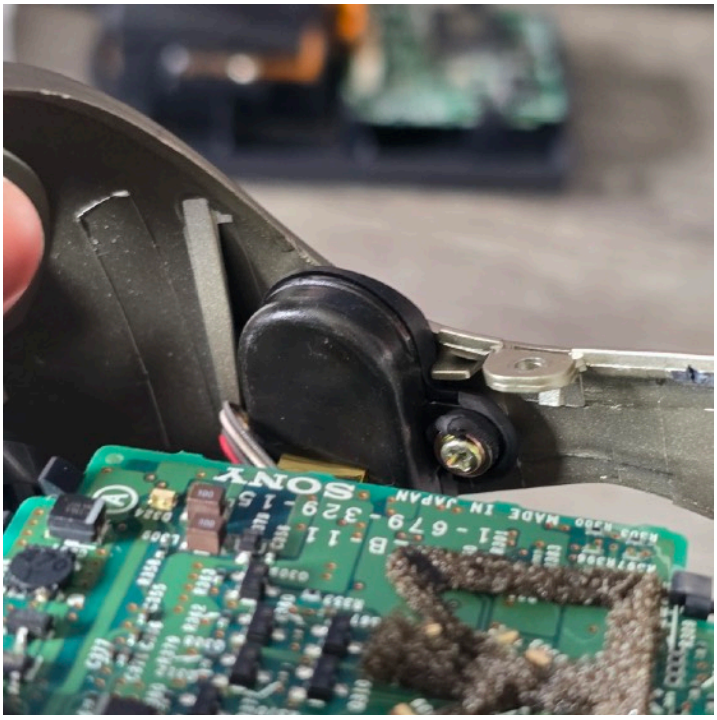
There are two pegs at the back that need unhooking, so tilt the coloured plastic backwards and off to unhook. There are wires connected to the coloured plastic, so don't try lifting the plastic away just yet - I'll teach you how to deal with that next!

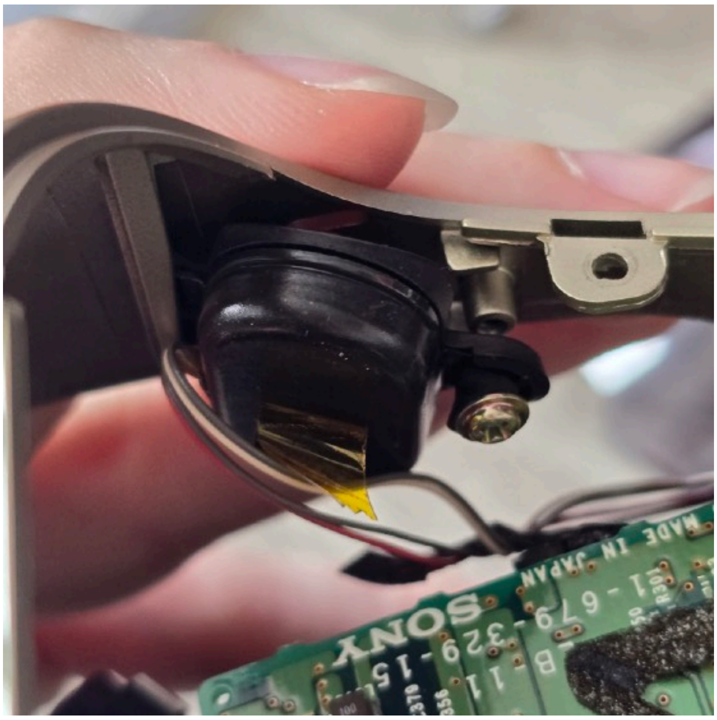
The black facemask is obstructing our view of underneath, so squeeze inwards from both sides and wiggle it out.

Next we will deal with the two microphones which are mounted either side of the inside of the coloured plastic.



When you lift off the black facemask you will be greeted by a green rectangular board with wires running into it. One each side you'll see a cable bundle composed of 4 thin wires going into a long rectangular connector that plug into the board. Some people like to disconnect these, but it's honestly not needed. Keeping the microphones plugged into the board and having them dangle is fine - it's never caused issues for me or people I know. If they get in the way, use masking tape to tidy them up while working in the head.





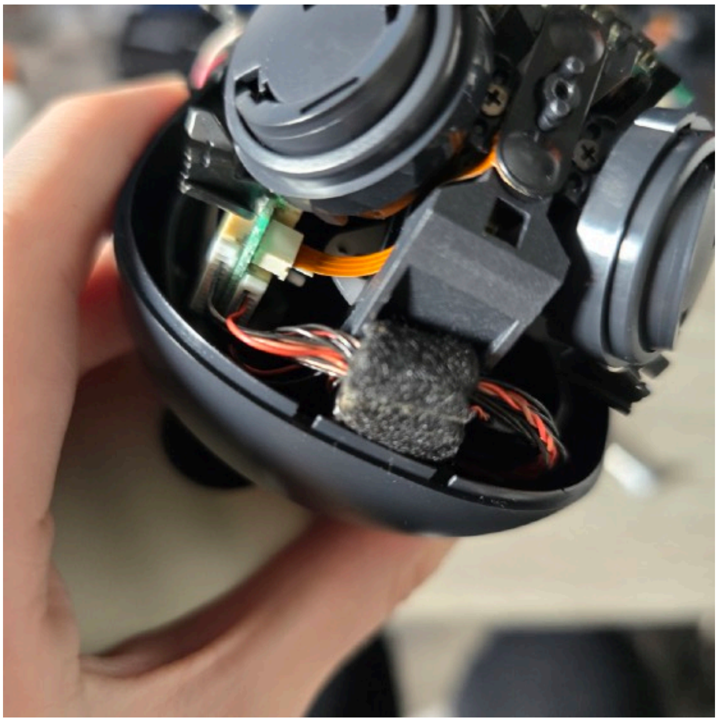
The microphones each side are held on by unique screws —>
Unscrew them and then pull the microphone forward and out of the notch.

Around each screw there is a rubber gasket that has aged quite badly and can make putting the screw back in a pain. Sometimes it even liquifies depending on storage conditions. If it is giving you issues it is fine to tear this part away, just make sure you leave the main rubber base under it that connects to the microphone in place. The screw will still securely hold the microphone to the plastic head shell.

Once the microphones are free, you can lift the coloured plastic off!

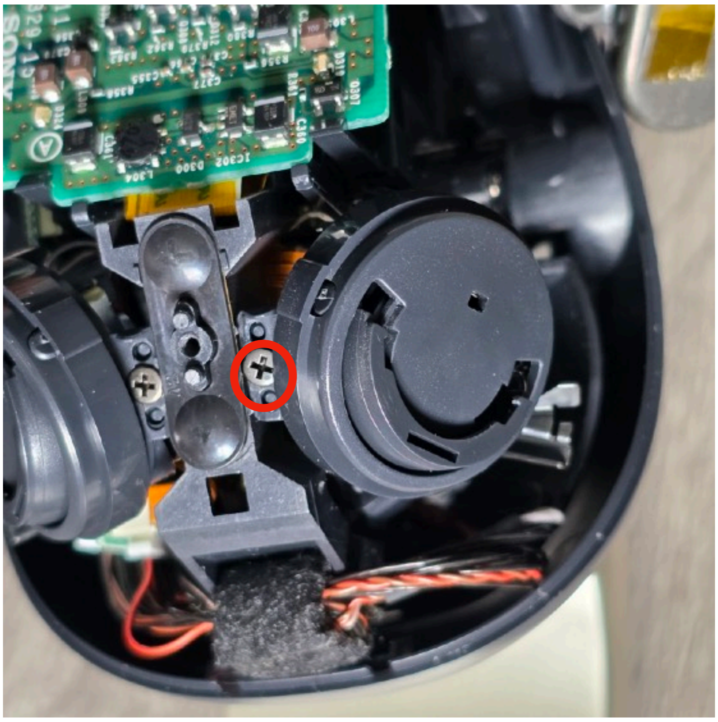
Top tip: If you see shielding tape over this board, it means your aibo has a Supercore head with the improved head clutches!

Next, grab this cable bundle by the fabric lump and gently free it from the ditch - it can sometimes be difficult to free it from this ditch area so if it doesn't immediately pop out, give it a bit of a wiggle while pulling up.
In the next steps we will deal with the cable bundle which will take a lot of time and patience. It can be extremely difficult as it will really fight you. You'll need the correct amount force while also being careful enough not to break anything.



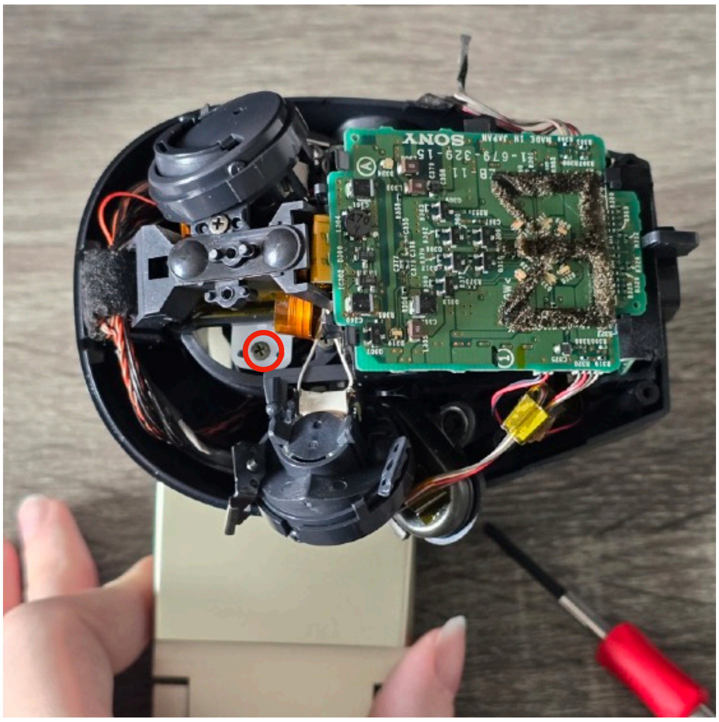
The wires may be different colours on your aibo to mine, but it's the same thing.

You'll notice it splits into two halves that plug into both sides of the board. We will deal with this later.



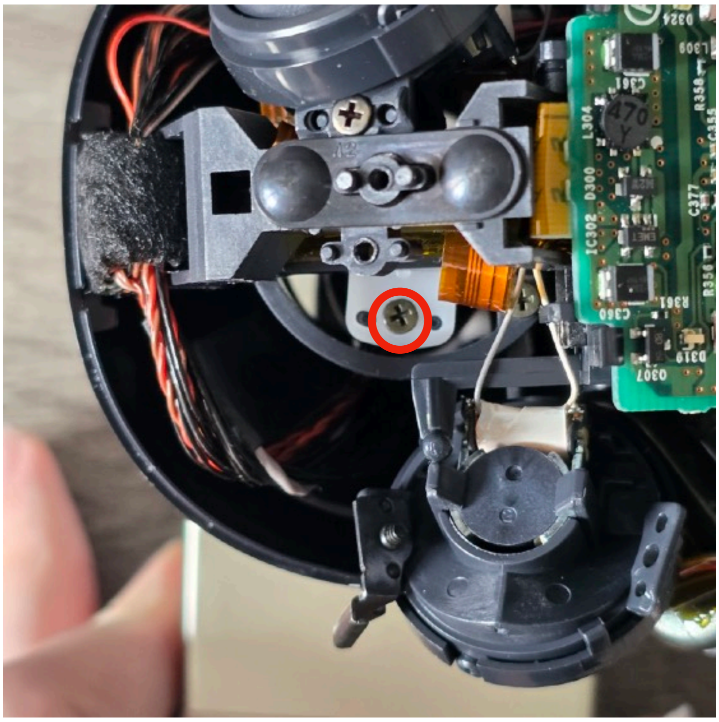
First lets unmount this ear hub. To do this, simply undo the screw on the top for this side.

You don't need to do both ears, just this one so we can get to a hidden screw underneath!



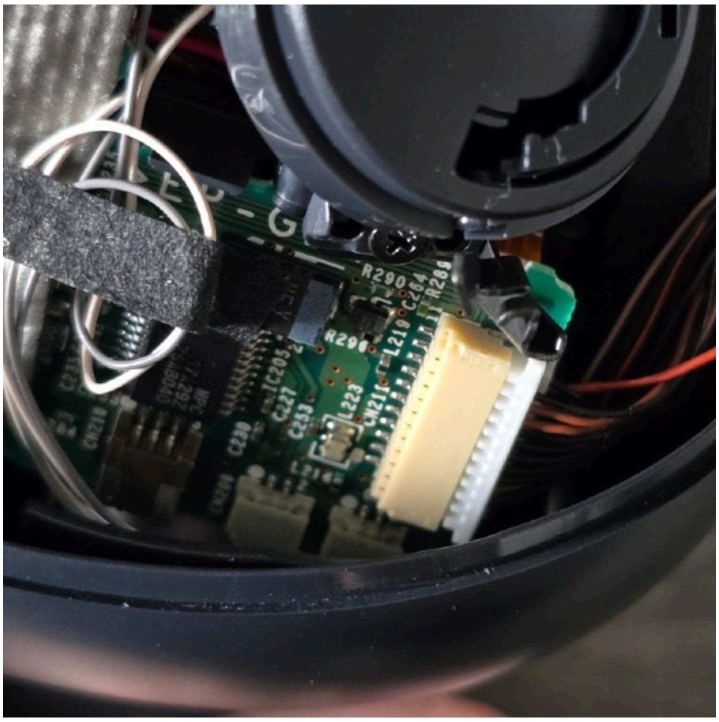
Once the ear hub screw is removed, gently lift the hub up and off the hook it is resting on.

Turn the Aibo's head in this direction to unveil the hidden screw.

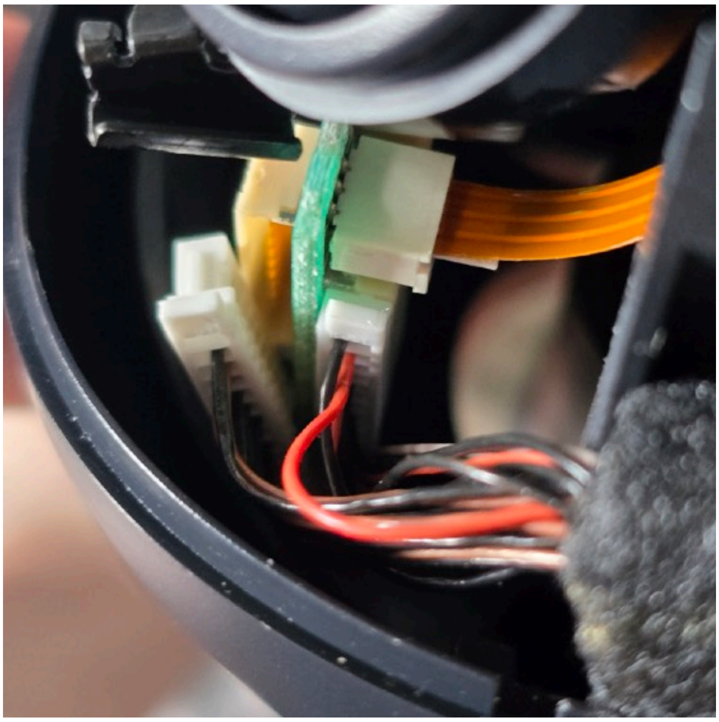


The screw in the white plastic part is what anchors the head to the body (yes really).

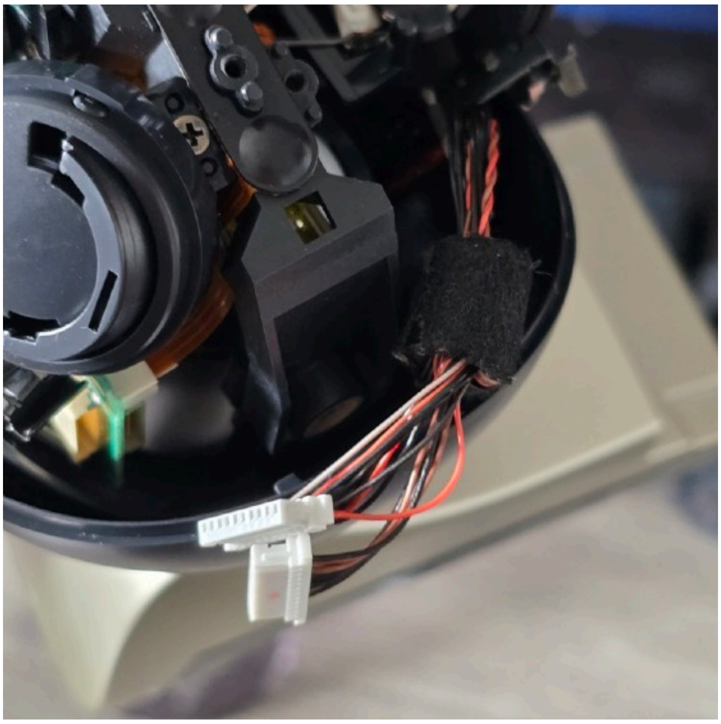
Take it out. To take the neck plastics off we have to decapitate the aibo by separating the two halves.



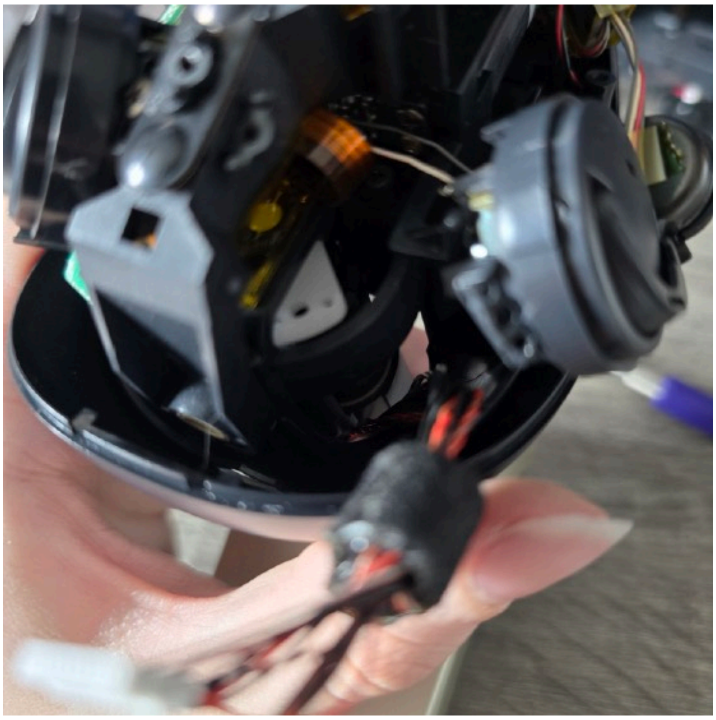
Now lets return to the cable bundle connectors I pointed out earlier… Using a tiny JIS flathead screwdriver or tweezers, gently push them out of the hubs. Put the screwdriver In these gaps and push the white connector out by alternating either side of the connector until it frees itself (its like when we wobble a connector out with our hands).



Do this for both connectors. Be extremely careful while taking these two out as the metal pins inside of the connector hubs can easily become bent or broken while ejecting/inserting them and you won't realise this until it's too late.



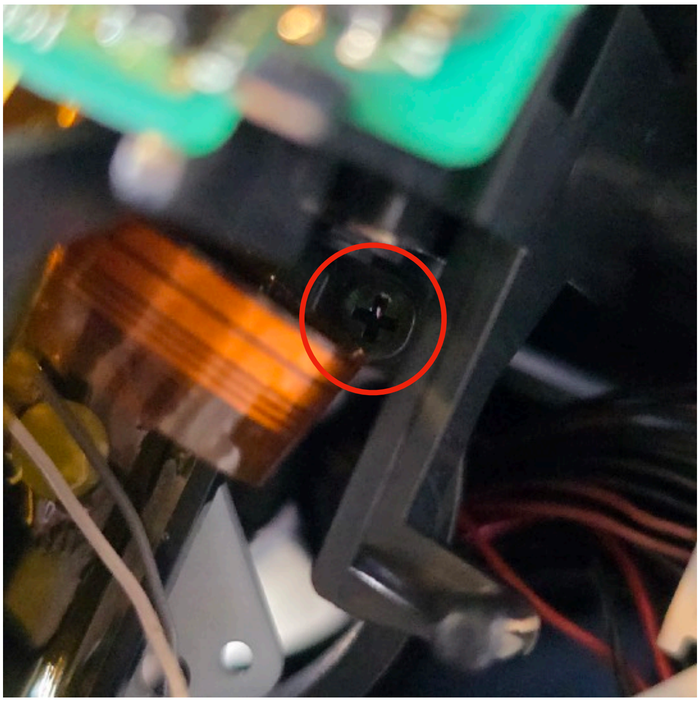
Grab the cable bundle and pull outwards.



Pull it around to the other side.

We need to guide and pull/push the cable bundle through this gap in the plastic under the head. This is quite difficult because Sony made the gap too small for the thick part of the bundle to fit through. So while pushing the bundle through, you also have to use your fingers to flex that left side plastic to widen the gap. You'll probably need to gently flex the servo column slightly to the other side too.

When you have the cable bundle in position, lower the neck part (which is now untethered from the head) so you can see what you are doing.

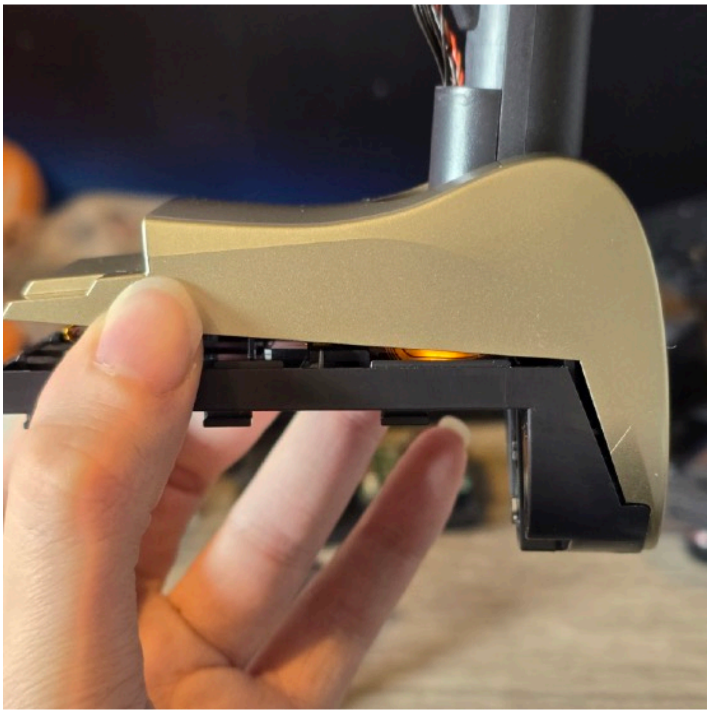


You can also try undoing this screw deep in the head, but it's not always guaranteed to help.

When you are getting close you will see the bundle lump poking through. Push it through hard from the other side while still flexing the plastic to widen the gap.

Once that part is through, gently guide the rest out.

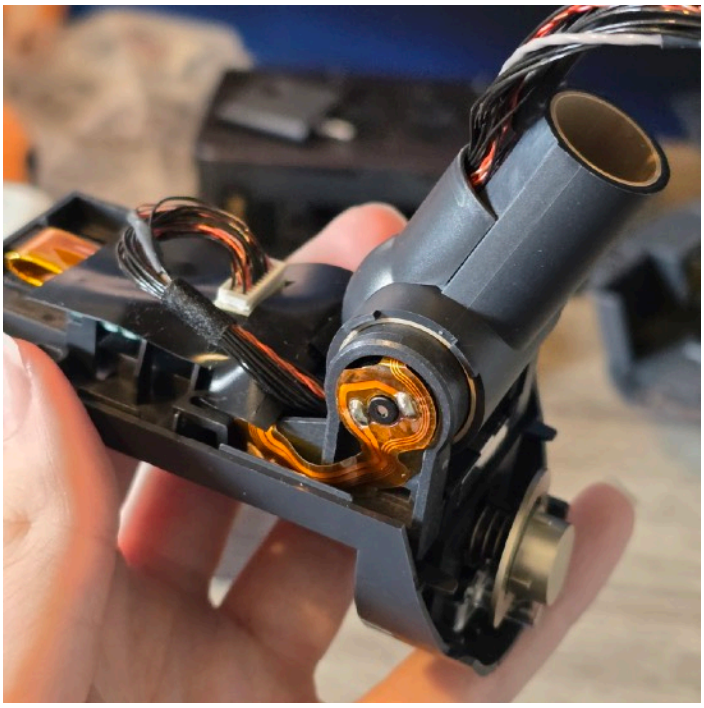
Congratulations - you survived the worst part of the ERS-210's design. Fun fact - they later rectified this in the ERS-220 by making the gaps bigger.



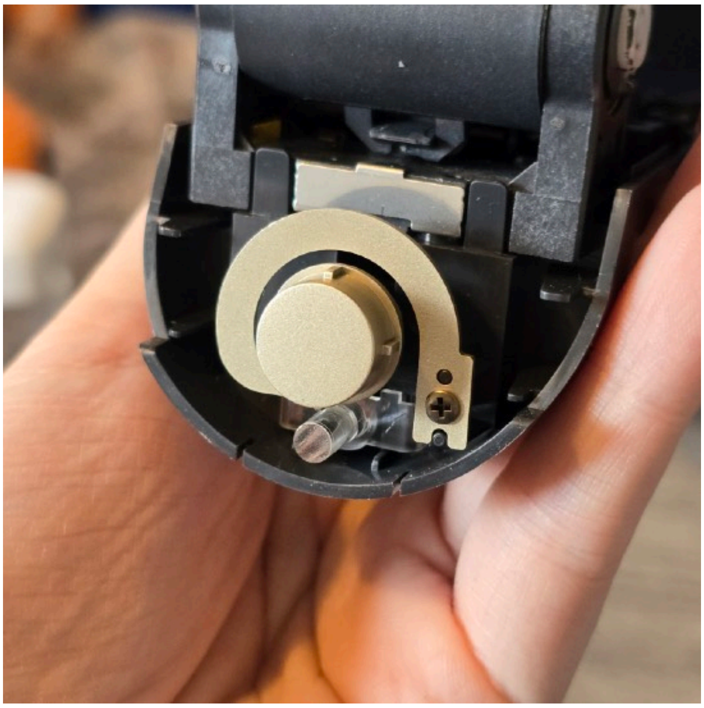
We can finally deshell the neck - take out all of the screws from the underside of the neck module. They are different lengths so please make notes of which ones go where.

Once removed, start lifting the coloured shell upwards from the back.

We need to get past the on button and light indicator plastic. Keep the button pushed in while pushing the neck plastic forward and then upwards so that it lifts up over the light indicator. Feel free to reposition the neck column to make this easier.



This is the bare internals of the neck area - this is where the servo is housed that causes DHS when it’s clutch assembly fails.



One last thing to do - unscrew the button and lift off.

Like mentioned before, make a note of where this spring came from so you don’t mix it up with the other spring from the head.